

Project Report

ICP(Ideal Customer Profile) Prediction Model

Domain: Machine Learning / Predictive Analytics / Customer Acquisition

Business Problem

Educational businesses receive a large number of student leads, but not every lead has the same probability of conversion. The objective of this project is to **identify the best potential customers**, prioritize high-intent leads, and help the sales team focus its efforts on students who are most likely to purchase.

Key Business Outcome

Identify the right customer for the right product at the right time — enabling data-driven lead prioritization and more efficient course sales.

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Project Type:

Machine Learning / Business Analytics Project

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1. Executive Summary

This project frames course sales as a binary classification problem: using information available before purchase, predict whether a lead will purchase a course within 30 days. The resulting model ranks prospects so sales capacity can be focused where it is most valuable.

79.5%	70.9%	51.0%	0.836
Accuracy	Precision	Recall	ROC-AUC

Key outcomes

- The model separates likely purchasers reasonably well, with ROC-AUC of 0.8362 and PR-AUC of 0.6951.
- At the default 0.50 threshold, precision is 70.9% and recall is 51.0%.
- The strongest business use is **lead prioritization**, not automatic rejection of low-scored students.
- Accounting and Finance show the highest observed purchase rates among the five course enquiries in the supplied EDA output.
- The prediction layer maps scores to Hot, Warm, Medium and Cold categories for operational use.

Interpretation

The model predicts purchase propensity; it does not establish causality. Thresholds should be aligned with sales capacity and the cost of missed versus unnecessary follow-ups.

2. Business Problem & Ideal Customer Profile

Education businesses can receive large volumes of enquiries from students with different levels of intent. Manual follow-up can spend too much time on weak prospects while highly engaged prospects wait too long. This project addresses that problem by converting observable lead information into an ICP score.

What the ICP represents

- **Profile:** age, education, occupation and city.
- **Intent:** course enquiry and lead source.
- **Behavior:** website visits, course/pricing page views, brochure download, email opens/clicks, WhatsApp responses and calls answered.
- **Engagement:** follow-up count, webinar/demo attendance and video watches.
- **Recency:** days since last activity.

Target definition

purchased_within_30_days is the binary target. Positive means the lead purchased within the 30-day window.

Leakage control

The training design explicitly avoids post-purchase information such as **purchased_course**. Only pre-purchase signals are included, which is essential for prospective scoring.

Business translation

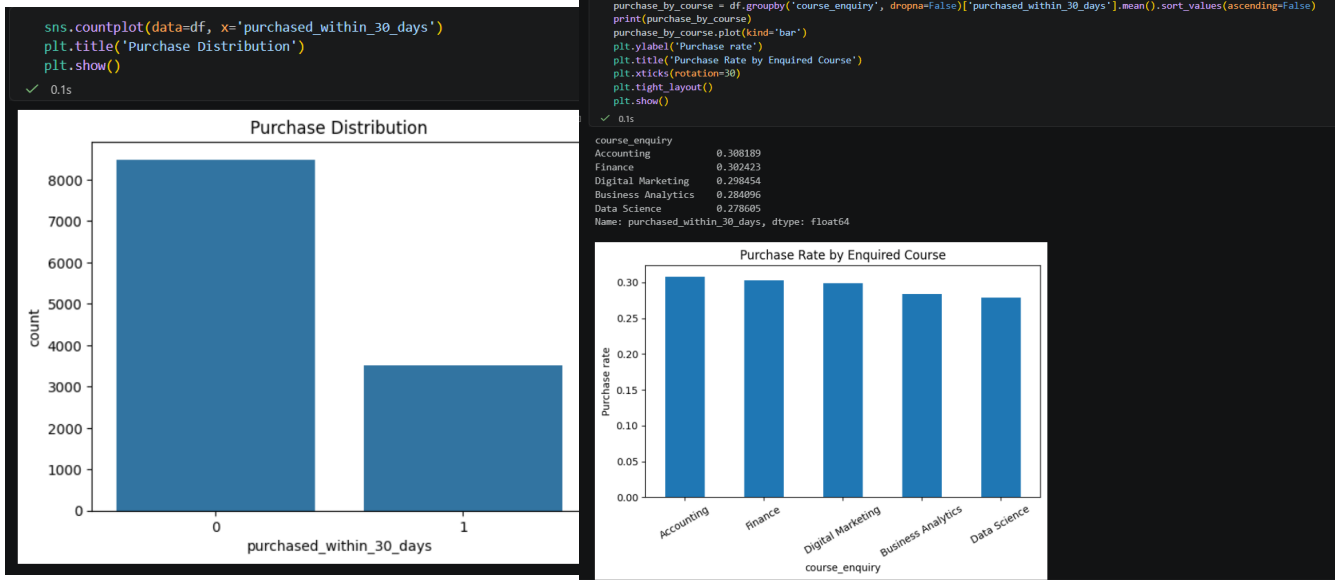
Model output	Business action
Hot (≥ 80)	Immediate counselor contact and personalized conversion action
Warm (60–79.9)	Priority follow-up and course-specific messaging
Medium (40–59.9)	Nurture and monitor engagement changes
Cold (< 40)	Lower-cost automated nurturing

3. Data Understanding & Exploratory Analysis

The project uses a supplied dataset of 12,000 student/course leads. The EDA notebook checks shape, missing values, target distribution, course-level purchase rates, behavioral differences, outliers and potential leakage before modeling.

Target distribution

The supplied chart shows a clear class imbalance: non-purchasers substantially outnumber purchasers. This makes precision, recall, F1 and PR-AUC important alongside accuracy.



Observed course purchase rates

Accounting: 30.82% • Finance: 30.24% • Digital Marketing: 29.85% • Business Analytics: 28.41% • Data Science: 27.81%.
The spread is modest, so course enquiry alone is not enough to define the ICP.

EDA takeaway

The final modeling strategy appropriately combines profile attributes with multiple engagement signals instead of relying on a single demographic or course field.

4. Features, Preprocessing & Leakage Control

The training pipeline separates numeric and categorical variables and applies preprocessing inside a scikit-learn Pipeline/ColumnTransformer. This keeps preprocessing consistent across validation, training and prediction.

Feature group	Examples	ICP role
Profile	Age, education, occupation, city	Lead context
Intent	Course enquiry, lead source	What the lead wants / acquisition route
Web behavior	Website, course and pricing views	Depth of product research
Communication	Email, WhatsApp, calls	Responsiveness and interest
Engagement	Follow-ups, webinar, demo, video	Strength of active consideration
Recency	Days since last activity	Freshness of intent

Preprocessing pipeline

- Numeric features: median imputation followed by standardization.
- Categorical features: most-frequent imputation followed by one-hot encoding with unknown categories ignored.
- Preprocessing is embedded with the classifier in one pipeline.

Validation design

An 80/20 stratified train-test split is used. Model comparison uses 5-fold StratifiedKfold cross-validation on the training set, preserving class proportions.

Why this matters

The same transformation logic is reused at prediction time, reducing training-serving mismatch and making the scoring workflow reproducible.

5. Model Development & Comparison

Three classifiers were evaluated with the same preprocessing pipeline. Cross-validation tracked ROC-AUC, PR-AUC, F1, precision and recall. Average precision (PR-AUC) was used as the tuning objective for the XGBoost candidate.

Model	CV ROC-AUC	CV PR-AUC	CV F1	CV Precision	CV Recall
Logistic Regression	0.834	0.695	0.638	0.550	0.759
Xgboost	0.823	0.675	0.573	0.669	0.502
Random Forest	0.825	0.673	0.623	0.610	0.637

Comparison insights

- **Logistic Regression** had the strongest CV ROC-AUC (0.834) and PR-AUC (0.695), plus the highest CV recall (0.759).
- **Random Forest** provided a middle ground between precision and recall.
- **XGBoost** was selected as the main candidate for tuning.

Tuning

RandomizedSearchCV evaluated 25 parameter combinations with 5-fold stratified CV and average precision scoring. The search varied number of estimators, tree depth, learning rate, subsampling, column sampling and minimum child weight.

Important note

The final test metrics in this report belong to the tuned XGBoost model. The report does not invent the best parameter values because those values were not included in the supplied evaluation output.

6. Final Model Performance

The tuned estimator was saved as **purchase_model.joblib** and evaluated on the held-out test set. With 12,000 total leads and an 80/20 split, the test set contains 2,400 leads.

Metric	Result	Business interpretation
Accuracy	0.7954	Overall share classified correctly
Precision	0.7089	Reliability of a positive prediction
Recall	0.5100	Purchasers captured at 0.50 threshold
F1	0.5932	Balance of precision and recall
ROC-AUC	0.8362	Ranking discrimination across thresholds
PR-AUC	0.6951	Precision-recall performance under imbalance
Brier score	0.1426	Probability prediction error; lower is better

What the numbers mean

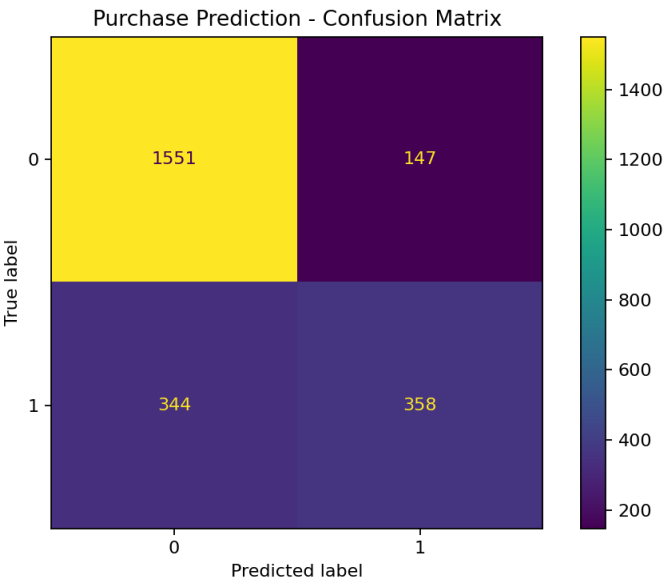
- ROC-AUC of **0.836** indicates useful separation between purchasers and non-purchasers across thresholds.
- At 0.50, **70.9% precision** means roughly 7 in 10 predicted purchasers were actually positive in the test set.
- **51.0% recall** means the default threshold captures about half of actual purchasers.
- The threshold can be changed when the business values coverage or precision differently.

Business conclusion

The model is best treated as a prioritization engine that helps sales teams decide where to spend scarce attention.

7. Confusion Matrix & Error Analysis

The held-out confusion matrix shows model decisions at the default probability threshold of 0.50.



Outcome	Count	Meaning
True Negative	1,551	Non-purchasers correctly classified
False Positive	147	Lead prioritized but did not purchase
False Negative	344	Purchaser missed by the 0.50 cutoff
True Positive	358	Purchaser correctly prioritized

Business interpretation

The largest opportunity is the **344 false negatives**: genuine purchasers not flagged at the default cutoff. If missing buyers is costly, the business can lower the threshold. If counselor capacity is very limited, a higher threshold can increase precision and reduce the number of leads contacted.

Key lesson

Model performance is a business trade-off, not just a technical score: the right operating point depends on contact capacity, lead value and the relative cost of false positives versus false negatives.

8. Lead Scoring, Threshold Strategy & Sales Workflow

The prediction layer accepts one lead record, generates a purchase probability, converts it to a 0–100 score, and assigns an operational category. This turns the ML model into a usable sales decision layer.

Score	Category	Suggested action
80–100	Hot	Immediate human follow-up; high-touch conversion action
60–79.9	Warm	Priority follow-up and personalized messaging
40–59.9	Medium	Nurture, monitor and re-score
0–39.9	Cold	Automated / low-cost nurturing

Threshold trade-off from supplied analysis

Threshold	Precision	Recall	F1	Leads selected
0.30	56.5%	74.8%	0.643	38.8%
0.40	63.8%	64.2%	0.640	29.5%
0.50	70.9%	51.0%	0.593	21.0%
0.60	76.6%	39.2%	0.518	15.0%
0.70	82.1%	26.8%	0.404	9.5%
0.80	86.5%	16.4%	0.275	5.5%

- **0.40:** 63.8% precision, 64.2% recall, selecting about 29.5% of leads.
- **0.50:** current default; 70.9% precision, 51.0% recall, selecting about 21.0%.
- **0.70:** 82.1% precision, 26.8% recall, selecting about 9.5%—useful when sales capacity is very limited.

9. Recommendations, Deployment & Conclusion

Recommended operating model

- **Rank first, contact second:** sort leads by predicted purchase probability and allocate human effort from highest to lowest score.
- **Use capacity-based thresholds:** do not assume 0.50 is permanently optimal; revisit the cutoff using counselor capacity, contact cost and conversion value.
- **Combine score with course context:** the model predicts propensity while EDA provides course-level purchase-rate context.
- **Re-score dynamically:** new website, email, call, webinar or demo activity can change intent.
- **Monitor outcomes:** track precision, recall, PR-AUC, conversion by score band and actual revenue after deployment.

Deployment flow

Lead data	Preprocessing	XGBoost	Probability	Score band	Sales action
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Conclusion

This project converts student lead data into a practical customer-acquisition decision system. The final tuned XGBoost model achieves **0.836 ROC-AUC**, with **70.9% precision** and **51.0% recall** at the default threshold. Its main value is helping the organization decide **which leads deserve attention first**.

Source artifacts used

train.py • predict.py • final_metrics.json • model_comparison.csv • threshold_analysis.csv • 01_data_understanding_eda.ipynb • supplied confusion matrix and EDA screenshots. The report intentionally avoids inventing values not present in the supplied outputs.

Final message: Find the best customer for the product, prioritize the right conversation, and use measured outcomes to continuously improve the ICP.